MIL-A-38331(USAF)
AMENDMENT 4
3 NOVEMBER 82
Superseding
Amendment 3
20 May 1977

MILITARY SPECIFICATION

AMPLIFIER-POWER SUPPLY AM-4080/ASN-55, LEVELING

This amendment forms a part of Military Specification MIL-A-38331, dated 3 August 1964 and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1: Under "SPECIFICATIONS, Federal", delete "QQ-P-416" and substitute the following:

"PPP-B-601 - Boxes, Wood, Cleated Plywood
PPP-B-636 - Box, Shipping, Fiberboard"

* 2.1: Under "SPECIFICATIONS, Military", delete "MIL-M-26512, MIL-R-26667, MIL-R-27542" and "MIL-D-70327". Add the following:

"MIL-P-116 - Preservation, Methods of
DOD-D-1000 - Drawings, Engineering and Associated Lists
MIL-C-83488 - Coating, Aluminum, Ion Vapor Deposited"

* Add the following boxed paragraph to bottom of page:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

2.1: Under "STANDARDS, Military", delete "MS33586" and add the following:

"MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-470 - Maintainability Program Requirements (For Systems and Equipment)
MIL-STD-781 - Reliability Tests: Exponential Distribution
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production
MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-889 - Dissimilar Metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods"

FSC 6610

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3.1: Delete paragraph entirely and substitute the following:

* "3.1 First article. When specified (see 6.4), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4).

PAGE 3

3.4.2.1, line 3: Delete "MS33586" and substitute "MIL-STD-889".

PAGE 4

3.5.3, line 4: Delete "0.33" and substitute "3.0". Line 5, delete "MIL-R-27542" and substitute "MIL-STD-785".

3.5.3.1, line 2: Delete: "MIL-R-26667" and substitute "MIL-STD-781".

PAGE 5

3.5.9,d: Delete and substitute the following:

"d. A 60-second timed, 115V 400 cps signal initiated by a ground pulse to control the fast erection cycle of an external displacement vertical gyroscope."

3.6,g: Delete "10 g's" and substitute "15 g's".

PAGE 6

3.8, line 1: Delete "MS17322-8" and substitute "MS17322-10".

3.10, line 3: Delete "figure 1" and substitute "figure 2". Under "Use", opposite pin J, delete "input" and substitute "output".

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Figure 1, pin J, delete "INPUT" and substitute "OUTPUT".

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Under "Use", opposite pin N, delete "input" and substitute "output".

PAGE 9

3.11.4, line 2: Delete "phase P" and substitute "phase B".

3.11.6: Delete entirely and substitute the following:

"3.11.6 Power consumption. The power consumption shall not exceed 15 va ac nor 5w dc (during starting or operating conditions)."

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Figure 2: Delete height dimension $\frac{\text{"3.543"}}{3.590}$ and substitute $\frac{\text{"3.653"}}{3.568}$. Delete title and substitute: "Amplifier-Power Supply Outline".

PAGE 11

* 3.16.2: Delete paragraph entirely and substitute the following:

"3.16.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, in accordance with MIL-C-83488, Type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the cognizant engineering activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

PAGE 12

* 4.2,a: Delete "preproduction" and substitute "First article".

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* 4.3.3,d, line 2: Delete "v" and substitute "V".

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* 4.4: Delete "Preproduction and substituted "First article".

4.4.1.1, lines 2 and 3: Delete "shall be selected at random from the first 50 produced on the contract or order and" and substitute "representative of the production equipment on the contractor or order shall be".

* 4.4.3, title and line 1: Delete "preproduction" and substitute "first article".

PAGE 17

4.5.3, line 3: After "7,700 hours", add "as specified in 4.6.20".

PAGE 19

4.6.3, line 2: Delete "4.3.4a and 4.3.4b" and substitute "4.3.4".

4.6.4.1.1, line 4: Delete "4.6.4" and substitute "4.6.4.1.3 and 4.6.4.1.5".

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4.6.4.2, line 2: Delete "4.3.3.1" and substitute "4.3.3,a".

4.6.5: Under "input Signal", delete "Per 4.3.3.2" and substitute "per 4.3.3,b". Also, delete "Per 4.3.3.3" and substitute "4.3.3,c".

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4.6.7,b: Under "Input Signal", delete "4.3.3.2" and "4.3.3.3" and substitute "4.3.3,b" and "4.3.3,c", respectively.

4.6.8,b: Under "Input Signal", delete "4.3.3.2" and "4.3.3.3" and substitute "4.3.3,b" and "4.3.3,c", respectively.

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* 4.6.13: Delete entirely and substitute the following:

"4.6.13 Temperature-altitude. The unit shall be subjected to a temperature-altitude test in accordance with MIL-STD-810, Method 504.1, procedure I, equipment category 5, except step 5 shall be conducted at 85°C for 4 hours; step 7 shall be conducted at 85°C; and steps 11 and 12 shall be at 80,000 ft. The unit shall then meet the individual tests."

PAGE 23

4.6.14,c: Delete "+72°C" and substitute "+71°C".

4.6.14.1,d: Under "Input Signal", delete "4.3.3.2" and "4.3.3.3" and substitute "4.3.3,b" and "4.3.3,c", respectively, and under "Output Voltage" delete "13.0" and substitute "11.0".

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4.6.14.2, line 2: Delete "4.6.15,a at +72°C". In subparagraph d., under "Input Signal", delete "4.3.3.2" and "4.3.3.3" and substitute "4.3.3,b" and "4.3.3,c", respectively.

*4.6.17.1: Delete entirely and substitute the following:

"4.6.17.1 Vibration failure. The vibration failure test shall be conducted in accordance with MIL-STD-810, method 514.2, procedure I, equipment category b.1, table 514.2-II except that figure 5 shall be substituted for figure 514.2-2 of that table. Power shall be applied during the test. At the conclusion of the test, the units shall be subjected to the individual tests."

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Figure 5: Delete and substitute new figure 5 attached.

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* 4.6.17.2: Delete entirely and substitute the following:

"4.6.17.2 Vibration endurance. The vibration endurance test shall be conducted in accordance with MIL-STD-810, method 514.2, procedure I, equipment category b.1, table 514.2-II except that figure 5 shall be substituted for

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figure 514.2-2 of that table. The frequency shall vary at a logarithmic rate from 5 to 500 Hz and back to 5 Hz in approximately 15 minutes in each of the three (3) axes. The equipment shall be operating during this tests but need not give specified performance. At the conclusion of the test, the units shall be subjected to the individual tests."

4.6.18: Delete "method 513" and substitute "method 513.2".

4.6.19.1: "delete "507" and substitute "507.1".

4.6.19.2: Delete "508" and substitute "508.1".

4.6.19.3: Delete "509" and substitute "509.1".

4.6.19.4: Delete title and substitute "Dust (fine sand).". Also change "method 510" to read "method 510.1".

4.6.19.5: Delete "506" and substitute "506.1".

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4.6.20.4, line 4: Delete "MIL-R-26667" and substitute "MIL-STD-781".

4.6.21: Delete entirely and substitute the following:

4.6.21 Maintainability program. A maintainability program shall be established in accordance with MIL-STD-470 for the purpose of achieving the maintainability goals of paragraph 3.5.4."

Add the following new paragraphs to Section 4:

"4.7 Packaging inspection.

4.7.1 Quality conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

4.7.2 First article. When the unit container is capable of serving as the shipping container, first article inspection and rough handling tests as outlined in section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction."

* Delete Section 5 entirely and substitute the following new Section 5:

"5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified.

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5.1.1 Level A.

5.1.1.1 Cleaning. Unit shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Unit shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each amplifier-power supply AM-4080/ASN-55, leveling, shall be packaged in quantity unit packs of one each in accordance with Method IIC of MIL-P-116. Each unit shall be overboxed in PPP-B-636 carton. Carton shall be large enough to allow for application of sufficient cusioning material between container and bag, of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the amplifier-power supply when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Units shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified.

5.2.1 Level A. Units packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, styles A and B, Class overseas unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, and of minimum tare and cube, consistent with the protection required.

5.2.2 Level B. Units packaged as specified in 5.1.1 shall be packed in weather-resistant class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 are applicable.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129."

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Add new paragraph to Section 6 as follows:

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"6.4 First article. When a first article is required, the item will be tested and inspected in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article."

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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